

Psycholinguistic Conspiracy Marker Extraction and Detection

1. Basics

Project topic: *Psycholinguistic Conspiracy Marker Extraction and Detection*

Team: *Racing cats*

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2. Elevator Pitch

With the advent of social media, the emergence of conspiracy theories has become frequent. Conspiracy theories can be used to spread fear and misinformation, which makes it important to differentiate fiction from the truth. We investigate the phenomenon of conspiracy in the context of large language models, focusing on identifying common textual patterns and detecting them in text-based digital media. Our goal is to produce new empirical findings that will contribute to the field and the detection of conspiracies.

3. Motivation

Misinformation is a big problem in today's world. One specific type of misinformation that stands out is conspiracy theories. A conspiracy theory is a piece of text that contains very radical conclusions which aren't built on a rational basis. An interesting problem regarding conspiracy theories in NLP is "Can conspiracy theories be characterized only by their semantic structure?" At first glance, it appears not, but there exists research such as (Fort et al., 2023) that makes this a promising path to take. This is also a relevant problem since conspiracy theories evolve according to the current state of the world. It would be very beneficial to produce a model that can classify a conspiracy theory only based on its semantic structure. Since we plan on using modern methods of LLMs, a question of pretraining data also arises. Will our model work better if we additionally pretrain it on a conspiracy theory corpus?

4. Gap

Analysis of the semantic structure of conspiracy theories has been a common problem in psychoanalysis. (Fort et al., 2023) addresses this problem by training the model on data in which topical words have been masked. The issue with this paper is the usage of outdated models such as BART and SVM. There is potential to build on their work by using LLMs. (Platt et al., 2022) is based on a similar idea. The goal of that paper is to differentiate between conspiracy theory text and more general misinformation text. Another interesting research direction deals with pretraining. (Levy et al., 2021) investigates GPT-2's ability to generate conspiracy text. They explain this is a consequence of conspiracy text being deeply rooted in the pretraining data. The problem with this paper is that it deals with the outdated GPT-2.

5. Research Questions & Contributions

NB:

While researching this subject, many possible research questions came to mind. Currently, we don't believe it's feasible to answer all of them in one semester. But, we would still like to write them here and hear some feedback.

RQ1:

How well do modern LLMs identify the semantic structure and specific language patterns of conspiracy theories? What if we fine-tune our model on texts where all context dependent entities are dynamically masked?

Contribution:

This will give us a better insight in conspiracy theory texts. We will determine if conspiracy theories follow specific forms that are independent of factual truth and that can be used to differentiate between critically written text and a conspiracy theory. We hope to achieve better results than (Fort et al., 2023). We would also like to try out different masking methods which they didn't try as they mentioned this in their conclusion.

RQ2:

How are different conspiracy theories related content-wise ?

Contribution:

This is also mentioned in the conclusion of (Fort et al., 2023). It's much easier to cite them, "Further research will need to investigate in more detail the inter-relatedness between different conspiracy theories. A better understanding of how they relate content-wise may allow us a better understanding of how to create a robust training set that can be used to detect conspiracy theories out of domain."

RQ3:

Should the pretraining set contain conspiracy theories?

Contribution:

We aim to determine if additional pretraining on harmful data such as conspiracy theories can be done in a safe way which also improves the model in terms of conspiracy theory detection. By pretraining the model on explicitly labeled conspiracies and non-conspiracies, the model could become better at recognizing harmful data. This could lead us to safer models and become an opening for future research in the field of AI safety.

RQ4:

How does learning conspiracy theories affect models performance on other tasks?

Contribution:

Our goal is to find out whether a model pretrained on conspiracy theories will adopt unexpected characteristics when answering prompts or trying to complete tasks. (?) notes a particular difference in language patterns used in conspiracy theory-related texts, for example a more

emotional tone of speaking. We aim to discover whether such patterns would become prevalent in the models output.

RQ5:

Can we additionally fine-tune the model to be resistant to sarcasm?

Contribution:

This seems like a hard problem so we think it would be interesting to tackle it. We haven't searched for current research on this topic, but we believe there has been some research regarding this and that there is still a lot of work to be done here.

6. Experimental Setup

Datasets. *We will be using LOCO or Language Conspiracy Corpus, introduced by (?). LOCO is an 88-million-token corpus composed of topic-matched conspiracy and mainstream documents harvested from 150 websites. LOCO includes an abundant ($N=237$) set of linguistic features, ranging from the document upload date to factual reporting annotations.*

Models & Baselines. *We will use Llama-3.1-8B as the baseline model for our paper. The Llama-3.1 is a collection of pretrained and instruction tuned differently sized models (8B, 70B and 405B) that input and output text only. All of models from the collection are Open-Source and well-documented since their release in 2024. making them well-suited for our research and further pretraining. We will have to take data contamination into account.*

Evaluation. *As RQ1 presents a very similar idea to (Fort et al., 2023), with some significant differences, we will be using the same metrics as them and comparing results between our paper and theirs. For RQ2 we plan to compare the internal representations of conspiracy texts and calculate their similarity. For RQ3 and RQ4 we will test our final model and the baseline model on a few different publicly available training datasets and compare results between the baseline model and further pretrained model. As for RQ5 we're not sure yet.*

General Methodology. *(1) Choosing a randomized subset of the LOCO dataset and splitting this into a smaller, randomized dataset and creating an 80/10/10 split of training/development/test data (Fort et al., 2023). (2) Determining maskable tokens by masking named entities and nouns, and then masking a percentage of the tokens. (3) Training Llama-3.1-8B with dynamic masking. (4) Extracting conspiracy markers by observing model prediction differences. (5) Evaluating the results for each RQ and concluding the research.*

Risks. *No experimental proof to support RQ1: As RQ1 relies of previous research from (Fort et al., 2023), it is possible we will not be able to reproduce their results. In that case we will aim to prove the opposite of what RQ1 aims to achieve and comment the difference between our findings and (Fort et al., 2023). Compute requirements : Llama-3.1 collection can be demanding for the GPU even at its smaller size (Llama-3.1-8B). If this proves to be true during our experiment, we will switch to a smaller model such as Mistral-7B-v0.1 that will not impact the RQ's greatly.*

7. Justification for API Credits

Use case. *API Credits would enable us to use a model such as GPT-4 as our baseline model. Using one of the best performing models currently available would allow us to better extract conspiracy markers and language pattern nuances in the conspiracy theory text overall positively impacting on our RQs.*

Alternative plan. *Not receiving API Credits will impact our RQs as we rely only on Open-Source models of a considerably smaller size and compute abilities. Our plan is to answer the RQs with the given resources (Llama-3.1-8B) and note in our research the possibility of our results being impacted by the models size and ability.*

References

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- Levy, S., Saxon, M., and Wang, W. Y. (2021). Investigating memorization of conspiracy theories in text generation. In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, pages 4718–4729.
- Platt, A., Brown, J., and Venske, A. (2022). Toward detecting conspiracy language in misinformation documents. In *Proceedings of the 2022 Computers and People Research Conference*, pages 1–4.